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Improving Nutrition Security in Low- and Middle-Income Countries and the Role of Animal-Source Foods

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Keywords

animal-source foods, undernutrition, nutrition security, low- and middle-income countries, LMICs

Abstract

Nutrition security is challenging in regions where resources are limited and food production is naturally constrained. In low- and middle-income countries (LMICs), undernutrition is high for many reasons, including lack of nutritional diversity and low high-quality protein content. Interest in the role of animal-source food (ASF) in reducing nutrition insecurity is increasing, as evidence from LMICs suggests that consumption of ASF is strongly associated with reduction in stunting, improved diet quality, and overall nutrition, particularly in early stages of life. We review the strengths and limitations of ASF consumption in terms of accessibility, safety, and nutritional benefits compared to non-ASF sources. We present a critical discussion on existing barriers to ASF consumption and its future directions in LMICs. Understanding the role of ASF in improving nutrition security in LMICs is crucial to optimizing public health, designing appropriate interventions, and implementing effective policy in resource-poor settings.

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1. INTRODUCTION

Nutrition security is defined as the ability to consistently access equitable, healthy, and safe food and consume sufficient calories to maintain a normal and active life (**https://www.usda.gov/nutrition-security**). Inability to access food and adequate nutrition leads to chronic hunger or undernourishment, which negatively impacts educational outcomes and economic productivity, consequently exacerbating a vicious cycle of ill health and poverty (1, 2). Latest estimates suggest that more than 700 million people globally currently experience hunger, whereas 600 million are projected to remain chronically undernourished by 2030, specifically in low- and middle-income countries (LMICs) where rates are already alarmingly high (3). In 2019, more than half of all children under 5 that were stunted, which means low height-for-age (64%) and wasted or low weight-for-height (75%), lived in LMICs (4). Despite continuous interventions, global efforts remain inadequate to achieve reduction targets in stunting and wasting by 2030 (4–7). The proportion of undernourished in South Asia and sub-Saharan Africa (**Figure 1**) is still higher than in the rest of the world and may continue to rise if food and nutrition security remain distant from those who need it most (3).

Various predisposing factors at the individual, household, and societal levels influence undernourishment in resource-poor settings (8, 9). In Malawi (8) and Bangladesh (10), for instance, poor maternal access to nutritional education and counseling during pregnancy and maternal exposure to chronic undernutrition were identified as immediate determinants of child undernutrition. In some countries in Asia and Africa, cultural and social taboos and personal preferences limit access to certain foods, specifically during lactation (11). Moreover, securing food and nutrition is challenging in regions where access to food sources is naturally constrained and exacerbated by the impacts of the global pandemic, conflicts, surges in food prices, and climate change (3, 12, 13). In a recent study on COVID-19 and food insecurity in Uganda, 50% of the sampled households reported an inability to eat healthy foods, and 42% were insecure about acquiring adequate food supplies during the pandemic-induced lockdown (14). LMICs, which often already grapple with access to equitable basic services, also have lower purchasing power and low mobility to avail sufficient and nutritious diets (10, 14). In typical LMIC diets, dietary diversity is habitually low, dominated by starchy foods and deficient in high-quality animal proteins (15).

Whereas animal-source food (ASF) consumption is more common in high-income countries (HICs), it remains low in LMICs for multiple interacting reasons, including availability and accessibility (16–19). However, a recent report shows that international agencies and governments are more actively promoting ASF in LMICs because of its benefits and impacts on nutrition and

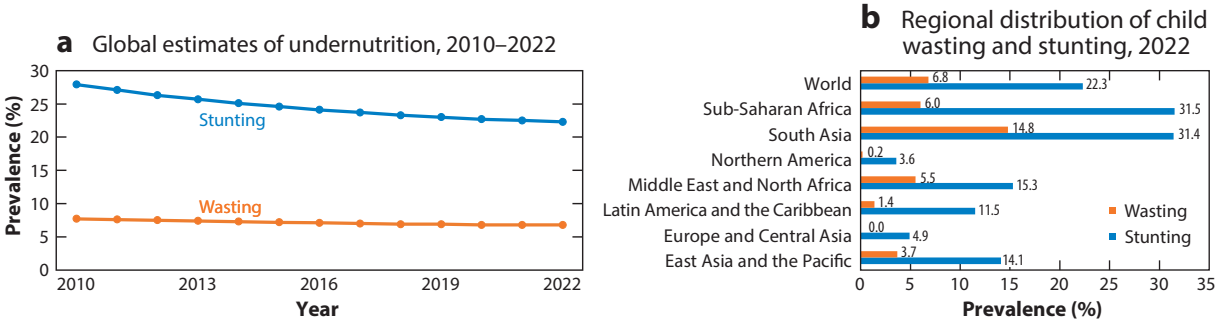

Figure 1

(*a*) Global prevalence of undernutrition (2010–2022) and (*b*) regional distribution of child wasting and stunting (2022; based on the UNICEF region classifications). Data from Reference 148.

health (20). Existing evidence suggests that ASF consumption in these contexts can play an important role in reducing stunting in children (21–23) and improving diet quality (24) and overall nutrition, particularly in early life (1, 12). Consumption of ASF during critical life stages, such as pregnancy or early childhood, is important, because ASF contributes higher quantities and bioavailability of essential amino acids, vitamins, and minerals (25, 26) that are necessary for proper brain development, increased immune response, and lowering of risk for complications (27–29). ASF offers valuable macro- and micronutrients that are often scarce in plant-based foods (26, 30). However, nutritional sciences support plant-source foods as healthier options than ASF because ASF consumption is associated with various noncommunicable diseases and chronic illnesses (26). Importantly, much of the literature on the health-related risk of ASF comes from HICs; thus, the distinct role of ASF in LMIC settings is an important consideration in understanding the literature (31). Some environmental advocates also support plant-based diets as more sustainable food options because of the contribution of ASF production (specifically livestock) to greenhouse gas (GHG) emissions that trap heat in the atmosphere, biodiversity loss, and environmental degradation (32). Other studies, however, argue that transitioning from ASF to plant-based diets may have potential nutritional consequences (e.g., decreased nutrients for bone health) if the alternatives are inadequate (30, 33). Environmental assumptions from ASF production also remain uncertain in some contexts (33) and may not necessarily apply to LMICs (34).

This article reviews ASF's contribution to nutrition security in LMICs by assessing its strengths and limitations in terms of accessibility, safety, and nutritional benefits compared to non-ASF sources. This review summarizes studies that focused on the relationship between ASF consumption and nutrition outcomes across the lifespan and presents a critical overview of the nutritional contribution of ASF, including the barriers to its consumption and future directions in LMICs.

2. NUTRITIONAL BENEFITS AND RISKS OF ASF CONSUMPTION

ASF, which are foods of animal origin—not only meat but also fish, eggs, milk, and milk products—have a high-quality protein index and ideal amino acid composition compared to plant-based sources (35). Even when consumed in small volumes, ASF can supply a sufficient amount of micronutrients, such as minerals (iron, zinc, calcium) and vitamins (B6, B12, D3), necessary for cell and tissue development that plant-based foods cannot provide in a typical serving (15, 35, 36). ASF creates a synergistic effect with other food and can be a viable remedy for micronutrient deficiencies, particularly in LMICs like Ethiopia (37), Zambia (38), and India (39) where undernutrition is prevalent and predominant dietary patterns are plant-based with little or no ASF.

2.1. Child Growth and Development

The nutritional contribution of ASF consumption to growth and the overall well-being of children has been reported in several studies in various low-income settings (15, 23, 40, 41). For example, in Kenya, a cohort of children under 5 from livestock-owning households who reported frequent egg (1 egg/day) and milk (1 ounce/day) consumption had significantly greater height gain within a 1-year follow-up period than those who did not (41). Similarly, in a clustered randomized controlled trial (RCT) in Burkina Faso, children whose mothers received a behavior change intervention package and who were fed on average 6 eggs per week had significantly improved z-scores for both underweight and wasting than the control arm, in which mothers received no training and children continued eating <1 egg per week (42). Z-scores (or standard deviations) are anthropometric values used to quantify distance of measurement (e.g., weight or height) from its reference mean and monitor child growth and nutritional status. Studies also support the introduction of ASF as a complementary food for children (aged 6–24 months), showing a positive

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and significant effect on both length- and weight-for-age z-scores (40, 43). Iron in meat and essential amino acids in eggs have excellent digestibility and efficient bioavailability, making them ideal components for complementary feeding of infants who have increasing nutritional requirements but immature gastric capacity and function (44–46). Recent World Health Organization recommendations emphasized that ASF, and eggs in particular, should be consumed daily by children 6–23 months old, because they have a high concentration of choline, a critical nutrient for neural development (47). Aside from eggs and meat, frequent consumption of small fish (at least 5 g/day) was also effective in marginally improving iron status in young children in Malawi (48). Moreover, milk has been highlighted as an important contributor to nutritional security, especially among young children (49). Studies using Demographic and Health Survey data from multiple countries concluded that dairy consumption leads to a significant reduction in the risk of stunting (23, 50).

Importantly, systematic reviews (1, 51, 52) have not found consistent high-quality evidence of any effect of ASF consumption on child growth outcomes, regardless of frequency, variation, or amount of ASF consumed (45). This varying evidence suggests possible confounding effects, potential errors in the previous review process, or an insufficient number of comparable studies to adequately assess the impact of ASF on child growth in LMICs (45, 51). The literature, however, points to the importance of ASF consumption in improving dietary diversity and in promoting optimal growth and early childhood development (24). A feeding intervention trial in Kenyan schoolchildren (mean age 7 years old) showed how academic performance, measured by test scores in key academic subjects, improved among those in the meat and milk groups compared to other feeding groups—plain githeri (a plant-based stew) and control (no intervention) (53). In Southern Ethiopia, children who consumed ASF also demonstrated better motor skills (e.g., running, kicking a ball, jumping) than those in the control group (54).

2.2. Pregnancy and Women's Health

ASF has also been documented to support micronutrient supplementation for women of reproductive age, especially during pregnancy and lactation (55). ASF contains iron, zinc, B vitamins, and folic acid, which are key nutrients for women to reduce maternal complications during pregnancy and compensate for losses during menstruation (25, 55, 56). ASF like red meat has more bioavailable forms of iron for anemia prevention than dark green leafy vegetables, which are also considered good sources of iron (56). A study in Bangladesh found that women who consumed ASF were less likely to be underweight or anemic compared to those who consumed no ASF (56). Consumption of ASF among women of reproductive age could improve the nutritional status of women and their future offspring and disrupt the intergenerational cycle of undernutrition (57). Another study reported inconclusive results on whether ASF consumption leads to better micronutrient absorption (58), thus warranting future investigation on ASF's role in pregnancy and lactation.

Despite nutritional benefits, the frequency of ASF consumption among women in LMICs remains low (25, 59). In Ethiopia (59), for instance, only 24% of the surveyed pregnant women consumed ASF at the time of the study. Though limitations are myriad and undoubtedly vary by context, ASF consumption among pregnant women in LMICs appears to be more likely when women are educated and/or own livestock (41, 42, 59). Livestock production and agriculture are pathways to increased household income, women's empowerment, and increased access to food and nutrition, particularly in LMICs. Livestock-producing households tend to consume more ASF produced within the household than those that lack that livestock holding, for example, milk from cattle-owning households and chicken or eggs from chicken-owning households (60–62). This has important implications for human well-being; where women are owners or co-owners

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of livestock, households, families, and children tend to benefit from improved food and nutrition security (63, 64).

2.3. Health of Aging Populations

As individuals age, their ability to absorb and utilize nutrients becomes less efficient; hence, it is critical to maintain a nutrient-dense diet at this stage (65). Adults and the aging population benefit from consuming ASF as they respond to their changing physical and physiological demands (25, 66). A recent Food and Agriculture Organization (FAO) report, which included studies predominantly from HICs, indicates that moderation in lean red meat intake (85–300 g/day) can protect older adults against iron deficiency (66) and improve muscle mass and function (67). Evidence from various reviews (66) also shows that consumption of eggs, milk, and dairy products among older adults reduces their risk for various noncommunicable diseases like diabetes, hypertension, and cancer, which may be contrary to our expectations, given literature coming from HICs (68). In China, a large-sample study of adults aged 20–80 years showed how physical and mental component scores of both male and female participants increased with greater ASF diet diversity scores. In Helsinki, an RCT among healthy adults found that replacing animal-source with plant-source proteins decreased iodine intake and vitamin B12 levels (69). Yet, there is a growing population of flexitarians among older populations—consumers who limit regular meat consumption (70). This shift is often motivated by the association between red meat consumption and age-related dysfunctions such as frailty (69), sarcopenia (66), and cardiovascular disease history (71). Red meat consumption, in particular, has been associated with increased renal hyperfiltration and metabolic complications, making plant-based proteins a potentially better option for those with existing chronic kidney disease (65, 72). The mechanisms of ASF and its impact on elderly health, however, are still unclear in LMICs, as most of the studies contain evidence from HICs (66).

3. PATTERNS OF ASF CONSUMPTION IN LMICs

Historically, per-capita consumption of ASF has been higher and more mainstream in HICs than in LMICs (17, 73). The annual average consumption in LMICs was only one-third of the meat and one-fifth of the milk consumed by HICs in the late 1990s (74). Between 1990 and 2018, global average ASF consumption was 8 kg/capita/year; however, in the United States, consumption was 124 kg/capita/year, whereas in African and South Asian countries (e.g., Sierra Leone, Rwanda, India, and Bangladesh), it was only 3–5 kg/capita/year, reflecting a strong bimodal pattern in global ASF consumption (73).

The rate of ASF consumption in LMICs is growing, however. This is likely due in part to the "livestock revolution"—a phenomenon in the 1970s wherein an increase in population, income, and urbanization fueled the increasing demand for animal products in low-income settings (74, 75). Recent FAO data show that trends in consumption of pork, poultry, eggs, and beef in LMICs has been increasing over the last 4 decades (76). As of the latest FAO recording, nearly 90,000 total metric tons of poultry, 81,000 metric tons of pork, 71,000 metric tons of eggs, and 43,000 metric tons of beef were consumed in LMICs in 2023 (**Figure 2**), and the consumption is expected to increase in the next years, especially in India, Pakistan, Philippines, Vietnam, and sub-Saharan Africa (75, 76). Other individual reports share the same trends, including distinctions in ASF consumption. In an Ethiopian study, consumption of dairy, meat, and eggs among young children increased from 31.3% in 2005 to 35.9% in 2011 and 41.5% in 2016, as behavioral campaigns promoting ASF consumption in this population and interventions for livestock production increased (37). In China, consumption of milk and dairy was higher among children (aged 4–17 years old) who lived

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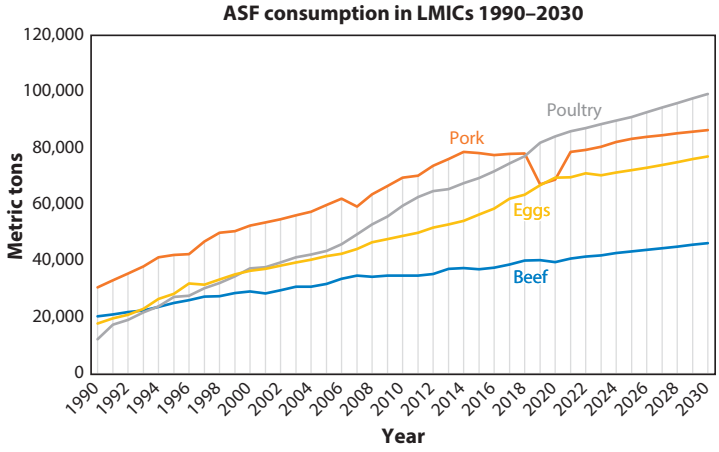

Figure 2

Animal-source food (ASF) consumption patterns in low- to middle-income countries (LMICs), 1990–2030. Data from Reference 76.

in highly urbanized area than those in rural communities, with consumption rates of 74% and 14%, respectively (77). Similarly, in India, household average per capita per month consumption of chicken (kg) and egg (n) was higher in urban areas and in states where poultry production was more accessible and much cheaper (78).

Globally, children and adolescents (<19 years old) with more educated parents and residing in urban areas consume more ASF (79). Yet, a systematic review emphasized that actual ASF intake is lower in infants and younger children, particularly in LMICs (79). In 1990–2018, actual consumption of dairy, milk, and yogurt was lowest in Nigeria, the Democratic Republic of the Congo, Bangladesh, and Tanzania, whereas intake from meat was significantly lowest in South Asia and sub-Saharan Africa (79). Although ASF is becoming more available in LMICs, many households still cannot afford to consume it regularly, like the case of chicken and egg in India (78). The growth of ASF consumption in LMICs is linked largely to income and, as such, remains inadequate to secure nutrition in the most vulnerable of low-resource settings. A review on resource economics reported that in most LMICs, increasing consumption of ASF, especially meat, has been associated with increasing gross domestic product, which is the opposite trend in many HICs (80). This decline in ASF consumption in HICs can be explained by personal preferences and rising consumer concerns regarding animal welfare, health, and environmental impacts of ASF that are yet to be explored in the context of LMICs (80). In most LMICs, however, plant-based diets are largely not chosen but a function of education, income, and access to affordable foods. In Timor-Leste, for instance, maternal ASF consumption tends to be higher in coastal areas than in mid-altitude areas, projecting consumption pattern variation based on the location and seasonality of available ASF for consumption (81).

Seasonality of ASF in LMICs can be impacted by various external shocks, such as extreme climatic events, conflict, and disease outbreaks. In East Africa, for example, foot-and-mouth disease outbreaks have been associated with decreased livestock sales and decreased household milk consumption among the affected households (82). Moreover, extreme climate events impact some ASF production systems in LMICs, such as that of pastoralism, which is particularly common in sub-Saharan Africa. Extreme temperatures, for example, pose threats to the availability of pasture and water and directly affect the health of livestock and livestock keepers (83). In Nigeria in 1983, a

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drought event combined with a rinderpest outbreak was responsible for hundreds of thousands of cattle deaths, with economic losses estimated to be around US$2 billion (84). Interventions that aim to increase ASF consumption in LMICs should also target small and large ASF producers to create environments resilient to shocks that threaten the supply of ASF. As the population of LMICs continues to grow, so does the need to increase access to ASF and mitigate the potential impacts of external shocks to ASF production, as it most often results in price increases, lowered productivity, and diminished ASF consumption at the household level.

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4. BARRIERS TO ASF CONSUMPTION

4.1. Sociocultural Barriers

Cultural beliefs may affect the restriction or inclusion of specific foods, the preparation of food, and the importance and meaning tied to the act of food consumption (85). They impact nutrient intake in several societies in LMICs and at specific stages of life.

4.1.1. Food taboos. Food taboos are strongly tied to cultural, religious, or spiritual influence and frequently are transmitted intergenerationally, which helps explain the persistence of food taboos in some cultural groups (85). Food taboos, which are most often restrictive, are frequently adopted during important periods of development, such as in pregnancy, the postpartum period, and early childhood (11, 34, 85, 86). For example, in areas of rural India, some pregnant women refrain from eating certain foods for their perceived influence on the child; some castes also restrict postpartum women's food intake to keep their stomachs empty (87). In Sudan, women reported avoiding some foods during pregnancy, including citrus and eggs, because of beliefs that these foods are harmful to their babies (88). In some areas of Kenya, pregnant women may avoid consuming eggs, meat, and milk, because they believe consuming these foods may lead to a complicated delivery (89). In southeastern Nigeria, seafood and egg consumption during pregnancy and for children under two years of age is restricted (90).

Religious norms and practices can also significantly shape diet practices, particularly for ASF (91–93). Meat is among the most commonly avoided food products, although the types of tabooed meat vary across cultural groups and traditions (94–97). For example, in most Indian religious and ethnic groups, beef consumption is forbidden (91). In Ethiopia, those who follow Orthodox Christianity refrain from consuming ASF, particularly pork, during fasting periods throughout the year (98). Additionally, those who follow Islam completely avoid eating pork. Although children and pregnant women can be exempt from some restrictions, religious practices of a given area may affect food systems and dietary norms of households and communities, thereby influencing children's and women's diets despite stated exceptions (99). Furthermore, fasting and other widespread cultural norms around ASF consumption can contribute to longer-term lower availability of and higher prices for ASF due to reduced investments and broader value chain effects (92).

4.1.2. ASF knowledge and perceptions. Nutritional knowledge among parents and caregivers, particularly knowledge on the nutritional content, safety, and adequate use of ASF, is also important for adequate ASF feeding practices (18, 86). Literature has recognized educational achievement and knowledge as important for ASF consumption, although education and knowledge are just one of many interrelated pathways toward improving ASF consumption and nutritional security and may not be sufficient on their own (100, 101). One study that analyzed preschool children from 56 LMICs found that maternal education has significantly stronger associations with various behavioral outcomes that can influence child nutrition compared to paternal education (102). An RCT designed to test the effects of a culturally tailored behavior change intervention to increase child egg consumption in Burkina Faso found that training and education

about the nutritional value of eggs facilitated a significant increase in child egg consumption (42, 101). Another RCT designed to increase child egg consumption in Ethiopia through gifting chickens to families and providing education on poultry production and the value of feeding eggs to children concluded that the provided education was a contributing factor to increased child egg consumption (103).

4.1.3. Intra-household food allocation. Dynamics within the household, such as intra-household food allocation, can also affect ASF consumption among household members. For instance, in some practices, women, and often children as well, are more likely to eat after men within the household, preventing women from both receiving adequate food and consuming nutrient-rich foods (such as ASF) (11, 104, 105). In South Asia (Bangladesh and India), many women reported eating the least or eating the least nutrient-dense foods (105). Similar results were found in women in the Indian Himalayas and Nepal (106, 107). In Nigeria, interviews with participants demonstrated that male household heads are often served more frequently and given more eggs than other household members (108). How food is allocated within households is complex and dynamic (109). Determinants of intra-household food allocation are influenced by individual household members' income, bargaining and decision-making power, social status, religion/ethnicity/caste, and local norms (110).

4.2. Economic Barriers

The barriers to ASF consumption also include its price, availability in the local market, and accessibility. For example, in an Ethiopian study, 99% of household surveys indicated that fish was not available in their local shops (111). In Bangladesh, mothers were three times more likely to feed ASF to children (8–23 months) if it was available at home (112). In some cases, livestock is viewed as a source of income rather than food, which limits its availability for local consumption. As reported previously, households that rear poultry and other livestock often choose to sell them instead of consuming them (113). Livestock is considered a valued asset, and households sell their livestock when necessary to cover other expenses like education, health-related fees, or ceremonies; households may also elect to sell livestock and buy frozen meat, as it is often less expensive than local meat (113).

In the last two decades, a significant increase in the real price of ASF has been noted. In Ethiopia (114), for instance, increases of 33% in the real price of beef, 36% in milk, and 32% in eggs were reported, even though Ethiopia has the largest livestock population in Africa (115). Eggs are also an exceptionally expensive source of calories in Africa, typically 10 times as expensive as starchy staples (116). A global analysis reports that the recommended diet, including ASF, is not affordable for most people in LMICs (117). Studies conducted in Bangladesh (118–120), sub-Saharan Africa (60), and Ethiopia (100) have similarly found the lack of affordability to be a significant barrier to ASF consumption. In many cases, households simply cannot afford meat, and plant-based food is more affordable than meat, eggs, and dairy (111). In other studies in Timor-Leste (113) and Kenya (121), the authors found that even though participants were aware of the nutritional benefits of ASF, they could not afford to consume it. Because ASF prices depend largely on production cost, quality, and seasonal availability of animal feeds (122), interventions to support small businesses, improve efficiency in transport and processing systems, and minimize food waste along the chain and during storage were suggested to reduce ASF prices (121).

Closely related to the cost of ASF, household access to and engagement in markets emerges as an important determinant of dietary diversity in LMICs (123, 124). The greater the access to markets, the higher dietary diversity and ASF consumption tend to be. A recent review underscores the important role of market access, particularly in large rural populations where agriculture is a

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dominant livelihood but market access is limited (124). Households' ability to sell ASF suggests a presence of viable markets, in which case, implementing a farm-level intervention may not be necessary to increase ASF consumption. Instead, income transfers or market-level interventions may achieve dietary improvements at a lower cost.

5. OTHER CONSIDERATIONS

5.1. ASF and Food Safety

The safety of consuming ASF vis-à-vis the quality of ASF is embedded in the various dimensions of animal production and food processing conditions (17). Many HICs, for instance, have reduced consumption of red meat because of its associated health risks, environmental pollution impacts, and animal welfare issues that are inarguably true in some contexts (125). Literature suggests that waste materials generated from ASF production, such as poultry litter and manure, often surpass the quantities required for fertilizing local agricultural land (126). Furthermore, livestock manure may harbor microorganisms and pharmaceutical residues, including antibiotics, which can contaminate the soil and water and contribute to zoonotic disease transmission and/or antimicrobial resistance (127). The use of antibiotics in ASF production (e.g., chickens, cattle, and pigs), particularly in highly industrialized farming systems, also poses a risk to human health and may outweigh the nutritional benefits from ASF (125, 128). Although antimicrobial residues rarely cause negative human health outcomes through consumption of ASF, there may be significant harm to human health if the use of antimicrobials in agriculture leads to resistance among zoonotic pathogens that then infect people (129). In addition, environmental enteric dysfunction, a subclinical inflammatory condition of the intestines characterized by reduced absorption and reduced barrier function, is associated with infection by enteric pathogens, including those commonly spread through livestock (130–132).

In a surveillance study (133) for *Escherichia coli* in dairy and meat products in Iran, pathogenic strains were detected in 45% of dairy and 52.5% of meat samples, but these findings were linked not to the ASF per se but to the process of pasteurization and inadequacy of sanitary conditions during ASF preparation, transport, and storage. Another investigation on marine pollution and fish consumption in Indonesia (134) reported that fish were the most frequently consumed ASF because of their perceived importance to nutrition yet simultaneously were regarded as a health risk given the concentrations of heavy metals and microplastics present in fish and aquatic ASF.

5.2. Environmental Impacts of ASF

Importantly, there are implications of ASF production on the environment that deserve consideration in efforts to increase ASF production and consumption. These include land use and soil health, water quantity and quality, and greenhouse gas emissions. Each of these is explored in more detail below.

5.2.1. Land use and soil health. Livestock farming produces ASF and requires large land areas for grazing and cultivating feed crops. Approximately 50% of global agricultural land and 20% of the earth's total land area are dedicated to livestock production (135). Yet, most of these grazing areas are often marginal lands that are not suitable for any purpose but grazing. The intensive use of land for monoculture crop cultivation for feed crops leads to soil erosion and degradation, with nearly 90% of cropland experiencing unsustainable soil loss (135). Despite these negative environmental effects, manure generated from ASF production has been used as a slow-release fertilizer for centuries (136). Evidence supports the positive impact of animal manure on soil fertility and various soil properties, ultimately contributing to overall soil health (25, 136). Studies

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have shown that manure application improves soil organic matter concentration and macro- and micronutrients, further improving crop yield (136, 137). Additionally, the application of animal manure modifies soil temperature, soil structure, and aggregate stability, which improves water infiltration and drainage (136). ASF production therefore may contribute to worsening soil health through extensive land use but could also improve soil health through manure generation.

5.2.2. Water quantity and quality. Extensive literature suggests that ASF production negatively affects water quantity, footprint, and quality (127, 138). Although livestock watering directly uses only 1.3% of total agricultural water use, the water required for cultivation and management of plants used primarily as feed for livestock dramatically increases water requirements. The daily amount of water dairy cattle require ranges between an estimated 21.8 L of water per animal and 127 L per animal; this could be higher depending on the farm's production system and the region's climate (127, 139).

5.2.3. Greenhouse gas emissions. ASF production is a major contributor to global climate change, contributing 12–18% of total GHGs, including carbon dioxide, methane, and nitrous oxide (135, 140, 141). An estimated 88% of emissions in the life cycle of animal products are generated on-farm, with only 12% derived from food system activities after agricultural production (e.g., processing, refrigeration, transport, retail, waste) (142). Studies conducted in Denmark and Canada revealed that diets high in ASF emitted more carbon dioxide compared to plant-based diets (143, 144). Although LMICs consume much less ASF than HICs, they are also the least efficient production systems, representing a disproportionate portion of enteric methane emissions compared to HICs and the sites of greatest potential gains through improved livestock management and production practices (145, 146). The intensification of livestock production, a necessary step for sufficient food production and food security in many LMICs, is also associated with increased vulnerability to antimicrobial resistance and zoonoses (147).

6. FUTURE PERSPECTIVES AND RECOMMENDATIONS

As summarized in the previous sections, existing evidence suggests that ASF contain nutritional values that are instrumental to the diets and economies of LMICs, yet their consumption remains low on account of formidable environmental, economic, and cultural challenges. ASF production contributes significantly to GHG emissions, as well as the degradation of soil and water quality and quantity. Traditional farming practices, alongside the expansion of industrial-scale agriculture, exacerbate these issues. To reduce environmental impacts, it is crucial to adopt sustainable livestock production systems that enhance production process efficiency while minimizing environmental harm. Sustainable intensification is a key strategy in achieving this goal. Sustainable intensification involves improved feeding practices that are nutritionally balanced and dense, including genetic improvement, culling of unproductive animals, fertility management, and decreased herd sizes (31). Economically, high market prices due to high production costs, limited access to markets, and limited infrastructure lower ASF consumption in LMICs. Targeted policy interventions to improve market access, improve value chains, and support smallholder farmers are crucial to bolster economic sustainability. On the health front, though ASF offers essential nutrients, it may contribute to diseases due to unsafe practices. Culturally tailored behavior change strategies and awareness programs that are embedded in local contexts can be beneficial in ensuring the preparation and consumption of ASF that is safe and sustainable and accounts for cultural beliefs, practices, and norms. Most existing research on risks of ASF consumption is based on populations from HICs, but there is also a need for furthering evidence-based research in LMICs that can highlight benefits and risks of ASF consumption among vulnerable populations. Such future research can

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help identify lever points in policy making for better practices in both ASF production (improving market access, sustainable livestock practices) and consumption.

7. CONCLUDING REMARKS

The primary objective of this review was to explore and summarize the potential contribution of ASF in nutritional security in LMICs as evidenced in literature and to identify the barriers responsible for low ASF consumption in these countries. ASF appears to be an important tool in the global effort to reduce undernutrition while simultaneously diversifying diets and increasing incomes of the most vulnerable. Because infants and young children have unique nutritional needs and very limited gastric capacity, they demand special consideration in ASF consumption, as density and bioavailability of nutrients from ASF prove crucial to child growth and development. Pregnant women and women of reproductive age, as well as the aging population, also have distinct energy and nutrient requirements to support physiological and physical demands. Moderate ASF consumption will be beneficial rather than detrimental for this specific population in LMICs. However, multiple challenges exist to scaling up ASF consumption in LMICs, including economic, environmental, sustainability, and cultural challenges. The review recommends policy and interventions aimed at promoting sustainable production systems, enhancing market access, ensuring food safety, and supporting sustainable diets through coordinated efforts involving governments, civil society, and international organizations. In conclusion, addressing the challenges related to low ASF consumption in LMICs requires a multifaceted approach that considers environmental, economic, and behavioral dimensions, ultimately aiming for sustainable development and nutritional security for future generations.

DISCLOSURE STATEMENT

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